

Analysis of Complex Sample Survey Data

SURV/SURVMETH 701

Lecture Notes: Module 2

**Complex Sample Weighting,
Survey Estimation and Inference**

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Three Elements of Design-Based Inference (CIs)

"....The form of this solution consists in determining certain intervals which I propose to call confidence intervals...." [Jerzy Neyman, 1934]

(1)		(2)		(3)
$\hat{\theta}$	$\pm$	$t_{df, 1-\alpha/2}$	$\cdot$	$se(\hat{\theta})$

where:

$\hat{\theta}$ = survey weighted estimate of θ ;

$t_{df, 1-\alpha/2}$ = critical value from Student t with df;

$se(\hat{\theta})$ = robust, design corrected estimate of $SE(\hat{\theta})$.

Weighting in Survey Analysis

Weighting is used to compensate for:

- Unequal probabilities of selection
- Nonresponse (typically, a unit that fails to respond)
- In post-stratification or calibration, to adjust weighted sample distributions for certain variables (e.g., age and sex) to make them conform to the known population distribution.

It is used to improve the accuracy (minimize bias) of sample estimates and to compensate for noncoverage and nonresponse.

Survey Weights

Weighting may simultaneously incorporate all three components: unequal probabilities of selection, nonresponse, and post-stratification

- 1) Weight for unequal probabilities of selection: w_{sel} ;
- 2) Weight for sample nonresponse: w_{nr} ;
- 3) Poststratification weight for population noncoverage and sampling variance reduction: w_{ps} .

Then compute the overall weight as:

$$W = w_{sel} \times w_{nr} \times w_{ps}$$

See: 1. Valliant, R., *The Effect of Multiple Weighting Steps on Variance Estimation*, *Journal of Official Statistics*, Vol. 20, No. 1, 2004, pp. 1–18.
2. Haziza, D. and Beaumont, J-F., *Construction of Weights in Surveys: A Review*, *Statistical Science*, Vol. 32, No. 2, 2017, pp. 206-226.

Sample Selection Weighting

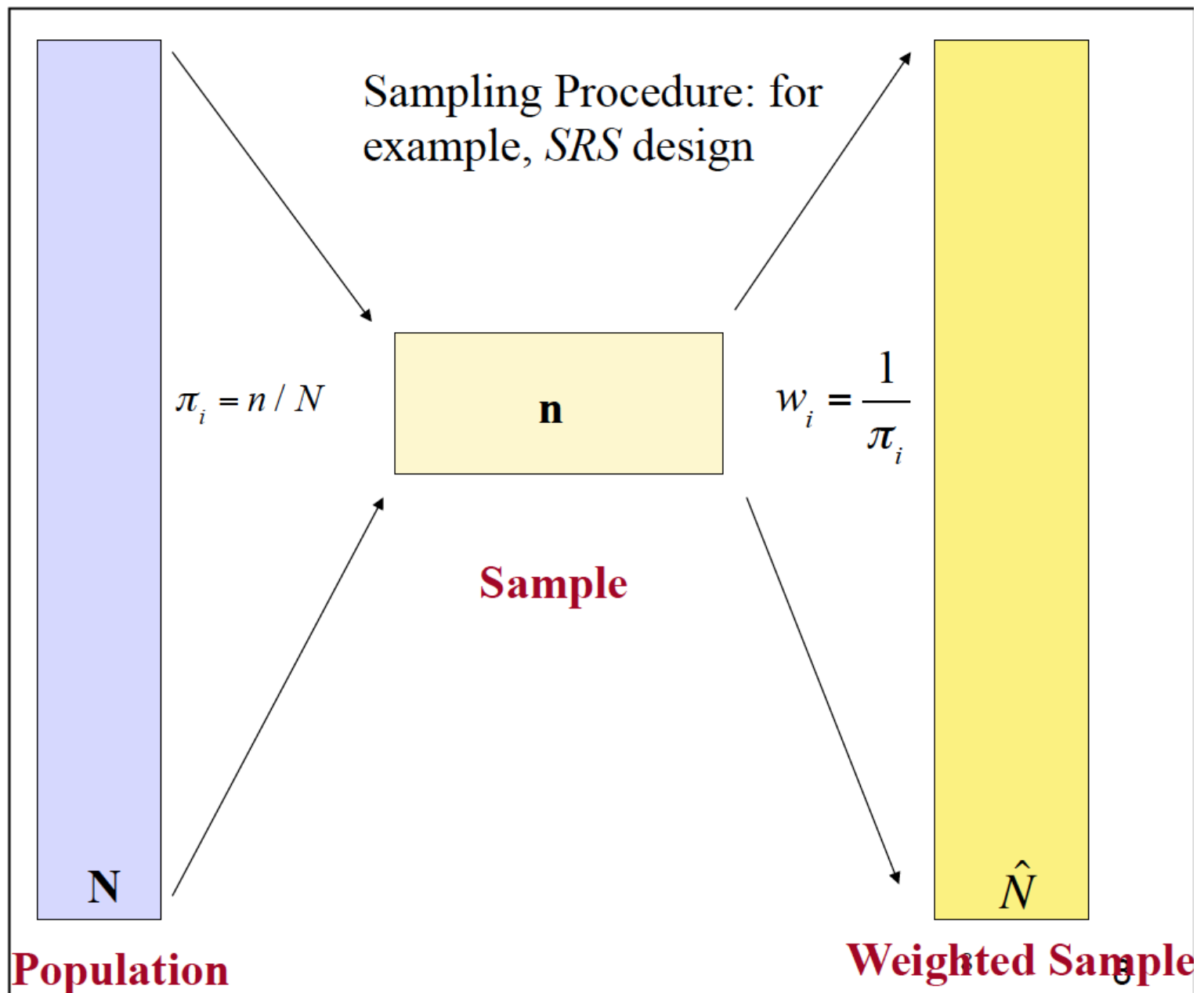
- Sample selection weights “map” the probability sample to the population that it represents.
- Suppose sample element i was selected with probability f_i . Then sample element i represents $(1 / f_i)$ elements in the population. That is, count the element i in the analysis by giving it a weight of:

$$w_{sel,i} = (1 / f_i).$$

- For example, a sample element selected with probability $1/10$ may be interpreted as “representing” 10 elements in the population.

Health and Retirement Study (HRS): Sample Selection Weight

Description of Sample Case						
Sample ID	Race/ Ethnicity of Respondent	Eligible Rs/ Household				
			$W_{sel, hh}$	$W_{sel, over}$	$W_{sel, elig}$	W_{sel}
A	Black	2	2000	1	2	4000
B	Black	1	2000	1	1	2000
C	Hispanic	2	2000	2	2	8000
D	White	1	2000	2	1	4000



epsem design

- epsem design: each population unit will have the same probability to be selected into the sample.
- Estimation under epsem design

- Population mean:

$$\bar{Y} = \frac{1}{N} \sum_{i=1}^N y_i = \frac{y_1 + y_2 + \cdots + y_N}{N}$$

- Sample mean:

$$\bar{y} = \frac{\sum y_i}{n} = \frac{y_1 + y_2 + \cdots + y_n}{1 + 1 + \cdots 1}$$

Non-epsem design

- But the selection probability may not be equal for every element
- Estimation: weighted estimator is required:

$$\bar{y}_w = \frac{\sum_i w_i y_i}{\sum_i w_i}$$

Example

- Suppose a sample of 20 households are selected from 2000 households in the population
- Within each selected household, one adult is selected
 - For 8 households, there was 1 adult – 3 of 8 adults reported recycling in the past month
 - For 6 households, 2 adults – 3 of 6 adults recycled
 - For 4 households, 3 adults – 3 of 4 adults recycled
 - For 2 households, 4 adults – 2 of 2 adults recycled

No. of households	No. of adults	Recycling
8	1	3 out of 8 adults
6	2	3 out of 6 adults
4	3	3 out of 4 adults
2	4	2 out of 2 adults

QUESTION: ONE ADULT SELECTED WITHIN EACH HH,
 1.WHAT IS THE PROB. OF SELECTING ADULTS? IS THIS
 DESIGN EPSEM?

2.WHAT IS THE PROP. OF ADULTS WHO RECYCLE?

Answer to Q1: Probability of selecting adults

- When 1 adult in the household, two stages of selection and

$$f_{sel,i} = (20/2000)(1/1) = 1/100 \quad w_{sel,i} = 100$$

- When 2 adults in the household,

$$f_{sel,i} = (20/2000)(1/2) = 1/200 \quad w_{sel,i} = 200$$

- When 3 adults in the household,

$$f_{sel,i} = (20/2000)(1/3) = 1/300 \quad w_{sel,i} = 300$$

- When 4 adults in the household,

$$f_{sel,i} = (20/2000)(1/4) = 1/400 \quad w_{sel,i} = 400$$

THIS DESIGN IS NOT EPSEM!

IDs	HH adult size	Recycling? (y=1/0)	Sampling Weight
1	1	1	100
2	1	1	100
3	1	1	100
4	1	0	100
5	1	0	100
6	1	0	100
7	1	0	100
8	1	0	100
9	2	1	200
10	2	1	200

IDs	HH adult size	Recycling? (y=1/0)	Sampling Weight
11	2	1	200
12	2	0	200
13	2	0	200
14	2	0	200
15	3	1	300
16	3	1	300
17	3	1	300
18	3	0	300
19	4	1	400
20	4	1	400

Answer to Q2: Weighted Estimation

- This can be represented in the weighted mean (proportion of adults who recycle) as

$$\bar{y}_w = \frac{\sum_i w_i y_i}{\sum_i w_i} = \frac{w_1 \cdot y_1 + w_2 \cdot y_2 + \cdots + w_n \cdot y_n}{w_1 + w_2 + \cdots + w_n} = \frac{\hat{Y}}{\hat{N}}$$

$$\bar{y}_w = \frac{100 \cdot 3 + 200 \cdot 3 + 300 \cdot 3 + 400 \cdot 2}{100 \cdot 8 + 200 \cdot 6 + 300 \cdot 4 + 400 \cdot 2} = 0.65$$

- The corresponding unweighted mean is

$$\bar{y} = \frac{\sum_i y_i}{n} = \frac{1+1+1+0+0+\cdots+1}{20} = 0.55$$

Exercise

- Suppose we draw a sample of 20 HHs from the list of 2000 HHs and 1 adult from each selected HH

Number of households	No. of adults	Response on Recycling?
6	1	1,1,0,0,0,0
6	2	1,1,0,0,1,0
6	3	0,1,0,1,1,1
2	4	1,1

Question: What is the proportion of the adults who recycle?

$$\bar{y}_w = \frac{\sum_i w_i y_i}{\sum_i w_i}$$

IDs	HH adult size	Response (y)	Sampling Weight
1	1		
2	1		
3	1		
4	1		
5	1		
6	1		
7	2		
8	2		
9	2		
10	2		

IDs	HH adult size	Response (y)	Sampling Weight
11	2		
12	2		
13	3		
14	3		
15	3		
16	3		
17	3		
18	3		
19	4		
20	4		

Number of households	No. of adults	Response on Recycling?
6	1	1,1,0,0,0,0
6	2	1,1,0,0,1,0
6	3	0,1,0,1,1,1
2	4	1,1

Question: What is the proportion of the adults who recycle?

What if not all the 20 adults in the sample responded?

Adjusting Weights for Nonresponse

- Not all selected sample elements will respond to the survey
- Even when observations are weighted to account for selection probabilities, differential nonresponse may result in biased estimates:

$$\boxed{Bias(\bar{Y}_R) = \bar{Y}_R - \bar{Y} = (1 - RR) \times (\bar{Y}_R - \bar{Y}_{NR})}$$

- RR is the expected population response rate
- Two common methods of developing adjustments for nonresponse:
 - Weighting class adjustment methods
 - Response propensity weighting adjustments

Weighting Class Adjustment for Nonresponse

- Form cells by cross-classifying respondents and nonrespondents based on known categorical variables that predict nonresponse and are associated with the variables of interest ($c = 1, \dots, C$ “weighting classes”).
- Assume probability of response within the cell is the empirical value of the response rate ($rrate_c$) for sample cases in cell c (per Kott, 2012, *Survey Methodology*, should be weighted!).
- Compute the nonresponse adjustment as the reciprocal of the response rate within each weighting class.

$$w_{nr,wc,i} = \frac{1}{rrate_c}$$

where : $rrate_c$ = the response rate for weighting class $c=1, \dots, C$

Example

- Suppose that not everyone in the sample of $n=1,000$ people responded

Class	$w_{sel,i}$	Sample size	Respondents	Income (y)
Education ≤ 12 yrs	1.00	500	400	\$60,000
Education > 12 yrs	1.00	500	450	\$80,000
Total		1,000	850	\$70,588 (v. \$70,000)

Question:

- a) Estimated Population Size=?
- b) Average Income=? Biased if ignore nonRspt's
- c) What are answers to a) and b) if base weight = 4.88 for edu>12 yrs?

Remedy: Weighting for Nonresponse - Weighting Class Methods

Step 1: Form the weighting class (done in the table)

Step 2: Compute response rate in each class

Step 3: Nonresponse weights are the inverse of the response rate in each class

- Assumption: data is missing at random (MAR) within classes

Weighting Classes	Respondents	$rrate_h$	$1/rrate_h$
Education ≤ 12 yrs	400	0.80	1.25
Education > 12 yrs	450	0.90	1.11
Total	850	0.85	

- Response rate in class h : $rrate_h$
- Nonresponse adjustment factor:

$$w_{nr,wc,i} = 1 / rrate_h$$

If all sample units respond (unbiased):

$$\bar{y}_s = \frac{\sum_{i \in s} y_i}{n} = \frac{60000 * 500 + 80000 * 500}{1000} = 70000$$

If nonrespondents are ignored (biased):

$$\bar{y}_r = \frac{\sum_{i \in r} y_i}{n_r} = \frac{60,000 * 400 + 80,000 * 450}{850} = 70,588$$

If nonrespondent adjusted (unbiased):

$$\bar{y}_r^w = \frac{\sum_{i \in r} w_{nr,i} y_i}{\sum_{i \in r} w_{nr,i}} = \frac{60,000 * 1.25 * 400 + 80,000 * 1.11 * 450}{1.25 * 400 + 1.11 * 450} = 70,000$$

Propensity Adjustment for Nonresponse

- Using variables that are known for respondents and nonrespondents, identify variables that predict nonresponse and are associated with the variables of interest.
- Model the propensity of response.
- Compute the nonresponse adjustment: 1) the reciprocal of the estimated propensity score; or 2) create weighting classes based on deciles of estimated propensity scores.

$$W_{nr,pro,i} = \left(\hat{p}_{resp,i} \right)^{-1} = \left\{ prob(response = yes \mid X_i) \right\}^{-1} = \left(\frac{e^{X_i \hat{\beta}}}{1 + e^{X_i \hat{\beta}}} \right)^{-1}$$

Therefore, we have Nonresponse Adjusted Weights:

$$w_{sel} \times w_{nr}$$

**What if
weighted sample distribution
 $\neq$
population distribution?**

Post-stratification Weighting

- The final step in survey weight development involves post-stratification of nonresponse adjusted weights to population controls:

$$w_{ps,l,i} = \frac{N_l}{\sum_{i=1}^{n_l} (w_{sel,i} \times w_{nr,i})} = \frac{N_l}{\hat{N}_l}$$

where :

$w_{ps,l,i}$ = the post-stratification weight factor for cases in post-stratum $l=1,\dots,L$; and

N_l = the population count in post-stratum l obtained from a recent Census, administrative records, or a large survey with small sampling variance.

- Some survey programs use more complex forms of post-survey adjustment to sample weights such as calibration, raking, etc.

Weighting for Noncoverage

- **Poststratification** is used to make the weighted sample distribution conform to a known population distribution

Example

Suppose that the sample distribution of *whether a household has children present* does not agree with the known distribution in the population

Children?	$\sum w_{sel,i} \times w_{nr,i}$	p_g	N_g	P_g
Yes	35,000,000	0.35	40,000,000	0.4
No	65,000,000	0.65	60,000,000	0.6
Total	100,000,000	1.00	100,000,000	1.0

Remedy: Weighting for Noncoverage - Weighting Class Methods

Step 1: Form the weighting class g (done in the table)

Step 2: Weighted sample distribution in each class p_g

Step 3: Population distribution in each class P_g

Step 4: Compute Poststratification adjustment factor

} NOT
EQUIVALENT

$$f_g = P_g / p_g$$

Children?	$\sum w_{sel,i} \times w_{nr,i}$	p_g	N_g	P_g	$f_g = P_g / p_g$
Yes	35M	0.35	40M	0.4	1.143
No	65M	0.65	60M	0.6	0.923
Total	100M	1.00	100M	1.0	1,000

In poststratification, the weights for the individuals in groups are adjusted up or down to obtain the distribution of the sum of weights that corresponds to the population distribution

A final weight

- The final weight is obtained after adjustments of the baseline weight for nonresponse and poststratification
- *Example (cont.):*

Education Children?	size of respondents	Final weight
Education ≤ 12 yrs		
Yes	140	$1 \times 1.25 \times 1.143 = 1.429$
No	260	$1 \times 1.25 \times 0.923 = 1.154$
Education > 12 yrs		
Yes	158	$4.88 \times 1.11 \times 1.143 = 6.195$
No	292	$4.88 \times 1.11 \times 0.923 = 5.003$
Total	850	Estimated popsize=(2,940)

HRS Final Weight

Description of Sample Case			W_{sel}	W_{nr}	W_{ps}	W_{final}
Sample ID	Nonresponse Adjustment Cell	Post-stratum				
A	Black, Northeast, Urban	Age 50-54, Male, Married	4000	1.3	1.04	5408
B	Black, South, Rural	Age 55-61 Female, Single	2000	1.15	.96	2208
C	Hispanic, West, Urban	Age 50-54, Male, Married	8000	1.25	1.06	10,600
D	White, Midwest, Rural	Age 55-61, Female, Single	4000	1.18	.97	4578

Regression Models of log-transformed Household Income for the HRS Black Subpopulation ($n = 2,465$).

Independent Variable		Regression Parameter Estimate [Standard Error, p -value]	
		Unweighted	Weighted (Design-based SE)
<i>Age (continuous)</i>		0.0026 (0.0058, $p = 0.66$)	0.0056 (0.0093, $p = 0.54$)
<i>Gender</i>	Female	-0.4629 (0.1246, $p < 0.001$)	-0.3034 (0.2199, $p = 0.17$)
	Male	Reference category	Reference Category
<i>Education</i>	Grade 0-11	-1.5585 (0.1991, $p < 0.0001$)	-1.9016 (0.2610, $p < 0.0001$)
	Grade 12	-1.0304 (0.2011, $p < 0.0001$)	-1.5177 (0.2871, $p < 0.0001$)
	Grade 13-15	-0.5145 (0.2152, $p < 0.0001$)	-0.7114 (0.1330, $p < 0.0001$)
	Grade 16+	Reference category	Reference Category
<i>Region</i>	Northeast	0.0804 (0.2743, $p = 0.77$)	0.1462 (0.1680, $p = 0.38$)
	Midwest	-0.3331 (0.2635, $p = 0.21$)	-0.2423 (0.2614, $p = 0.36$)
	South	-0.2525 (0.2476, $p = 0.31$)	-0.3519 (0.2405, $p = 0.15$)
	West	Reference Category	Reference Category
<i>Urbanicity</i>	Urban	-0.0697 (0.1690, $p = 0.68$)	-0.0553 (0.3751, $p = 0.88$)
	Suburban	0.5878 (0.1965, $p = 0.76$)	-0.0262 (0.4764, $p = 0.58$)
	Rural	Reference Category	Reference Category

A Simple Model for Losses In Precision due to Weighting

L_W = Loss due to weighting = Proportionate Increase in Variance of a Mean

$$\sim CV^2(W_i) = \frac{\frac{\sum_1^n W_i^2}{n}}{\left(\frac{\sum_1^n W_i}{n}\right)^2} \cdot n - 1$$

Subgroups:

$$L_{W,sub} \sim \frac{\frac{\sum_1^m W_i^2}{m}}{\left(\frac{\sum_1^m W_i}{m}\right)^2} \cdot m - 1$$

Generally, oversampling for subgroups: $L_{W,SUB} < L_W$ (see Kish, 1965)

Example of Weighting Loss in Disproportionate Sampling

Stratum	Weight	% of Hispanic Sample	Sample size
1	4	7%	70
2	2	17%	170
3	1.33	26%	260
4	1	50%	500
		100%	1000

For $n = 1000$,

$$L_w = \frac{\sum_{i=1}^n W_i^2}{\left(\sum_{i=1}^n W_i\right)^2} \times n - 1$$

$$= \frac{2759.9}{2,148,569} \times 1000 - 1$$

$$= .284$$

Survey Weighted Estimates (1)

$$\bar{y}_w = \frac{\sum_{i=1}^n W_i \cdot y_i}{\sum_{i=1}^n W_i} \text{ estimates } \bar{Y};$$

$$s_w^2 = \frac{\sum_{i=1}^n W_i \cdot (y_i - \bar{y}_w)^2}{\sum_{i=1}^n W_i - 1} \text{ estimates } S^2 ;$$

$$b_{1,w} = \frac{\sum_{i=1}^n W_i \cdot y_i \cdot x_i}{\sum_{i=1}^n W_i \cdot x_i^2} \text{ estimates the simple linear regression coefficient, } B_1.$$

Survey Weighted Estimation: More Complex Pseudo-Maximum Likelihood for Logistic Regression

Pseudo $\ln(\text{likelihood})$:

$$= \sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{i=1}^{n_{\alpha}} w_{h\alpha i} y_{h\alpha i} \ln(\pi(x_{h\alpha i})) \\ + \sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{i=1}^{n_{\alpha}} w_{h\alpha i} (1 - y_{h\alpha i}) \ln(1 - \pi(x_{h\alpha i})),$$

where

$$\pi(x_i) = \frac{e^{x_i B}}{1 + e^{x_i B}}, \hat{\pi}(x_i) = \frac{e^{x_i b}}{1 + e^{x_i b}}$$

and b = the vector of coefficient estimates that solves

$$U(B) = \left. \frac{\partial \ln L(B)}{\partial B} \right|_{B=b} = 0 \\ = \sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{i=1}^{n_{\alpha}} w_{h\alpha i} x'_{h\alpha i} y_{h\alpha i} - \sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{i=1}^{n_{\alpha}} w_{h\alpha i} x'_{h\alpha i} \hat{\pi}(x_{h\alpha i})$$

Scaling of Survey Analysis Weights

- Survey weights are generally released on a “population scale”:

$$\sum_{i=1}^n W_i \approx N \text{ (the population total)}$$

- Historically, many data producers scaled the survey analysis weights to sum to the sample size for the survey (NCS-R example above):

$$W_i^* = W_i \cdot \frac{n}{\sum_{i=1}^n W_i} \Rightarrow \sum_{i=1}^n W_i^* = n$$

- Weighted analyses, with the exception of estimates of population totals, are invariant to any linear scaling of the weights:

$$e.g. \ W_i^* = a \cdot W_i \ \text{or} \ W_i^* = W_i / b$$

Examples of Survey Weight Distributions

	NCS-R: NCSRWTLG	NCS-R: NCSRWTSH	NHANES: WTMEC2YR	NHANES: WTINT2YR	HRS: KWGTR	HRS: KWGTHH
<i>n</i>	5,692	9,282	5,563	5,563	18,467	18,467
Sum	5,692	9,282	217,700,496	217,761,911	75,540,674	82,249,285
Mean	1.00*	1.00*	39,133.65	39,144.69	4,144.73	4,453.85
SD	0.96	0.52	31,965.69	30,461.53	2,973.48	3,002.06
Min	0.11	0.17	0**	1,339.05	0**	0**
Max	10.10	7.14	156,152.20	152,162.40	16,532	15,691
Pctls.						
1%	0.24	0.36	0	2,922.37	0	0
5%	0.32	0.49	2,939.33	4,981.73	0	1,029
25%	0.46	0.69	14,461.86	16,485.70	2,085	2,287
50%	0.64	0.87	27,825.71	28,040.22	3,575	3,755
75%	1.08	1.16	63,171.48	62,731.71	5,075	5,419
95%	2.95	1.85	100,391.70	96,707.20	10,226	10,847
99%	4.71	3.17	116,640.90	113,196.20	12,951	14,126

Constructing Design-based Confidence Intervals: Degrees of freedom for the reference distribution

- Degrees of freedom are the number of independent comparisons (generally squared differences) which can be made between the elements of the sample.
- For test statistics (t, X^2 , F) that require estimates of variation in the data, df are related to the number of independent contrasts available to estimate the required variance(s)
- Consider:

$$t_{n-1, SRS} = \frac{(\bar{y} - \bar{Y}_0)}{se(\bar{y})} = \frac{(\bar{y} - \bar{Y}_0)}{\sqrt{s^2 / n}} = \frac{(\bar{y} - \bar{Y}_0)}{\sqrt{\sum_{i=1}^n [(y_i - \bar{y})^2 / (n-1)] / n}}$$

note: that for an independent sample of size n, there are n-1 independent contrasts to estimate s^2 , the unknown component of $se(\bar{y})$. $\therefore df=n-1$

Degrees of Freedom in Variance Estimation for Complex Sample Data

Leads to simple Rule:

degrees of freedom = # of clusters – # of strata

$$= \sum_{h=1}^H a_h - H$$

ex: Two cluster per stratum design.

$$\text{d.f.} = 2 \cdot H - H = H$$

See: Valliant, R. and Rust, K.F., *Degrees of Freedom Approximations and Rules-of-Thumb*, *Journal of Official Statistics*, Vol. 26, No. 4, 2010, pp. 585–602. (They propose a simple estimator of degrees of freedom that leads to improved confidence interval coverage relative to the simple rule above, which is currently used by most software packages.)

Degrees of Freedom in Confidence Interval Construction

$$CI_{.95} = \hat{\theta} \pm t_{.975, df}^* se(\hat{\theta})$$

$$t_{.975, 1} = 12.706$$

$$t_{.975, 5} = 2.5706$$

$$t_{.975, 10} = 2.2281$$

$$t_{.975, 20} = 2.0860$$

$$t_{.975, 30} = 2.0423$$

$$t_{.975, 40} = 2.0211$$

$$t_{.975, \infty} = 1.9600$$

$$Z_{.975} = 1.9600$$